

Experiment No.	5
Experiment Title	Decision Tree and Random Forest: A Comparative Classification Study
Student Name	Harshita B
Register Number	3122247001021
Faculty	Dr. Poreddy Ajay Kumar Reddy
Due Date	16th August 2026

Institution: Sri Sivasubramaniya Nadar College of Engineering, Chennai

Programme: M.Tech (Integrated) Computer Science & Engineering

Semester: V

Subject: ICS1512 & Machine Learning Algorithms Laboratory

Academic Year: 2026–2027 (Odd)

Batch: 2024–2029

Dataset: Wisconsin Diagnostic Breast Cancer Dataset

1 Objective

- To implement a Decision Tree classifier.
- To extend the Decision Tree into a Random Forest ensemble model.
- To study the impact of hyperparameters on overfitting and generalization.
- To select optimal hyperparameters using 5-Fold Cross-Validation.
- To compare single-tree and ensemble-tree models.

2 Problem Statement

The task is to classify observations from the Wisconsin Diagnostic Breast Cancer dataset into malignant and benign classes. A Decision Tree and a Random Forest classifier are trained, tuned using stratified 5-fold cross-validation, and evaluated on a held-out 20% test set. The experiment studies tree complexity, ensemble hyperparameters, overfitting, stability, and generalization.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC) Dataset
Dataset Source	UCI Machine Learning Repository / scikit-learn dataset copy
Number of Samples	569
Number of Features	30 numerical attributes
Number of Classes	2: Malignant (M) and Benign (B)
Missing Values	None in the supplied dataset

Train-Test Split	80% training, 20% testing; stratified
------------------	---------------------------------------

4 Exploratory Data Analysis

The notebook performs dataset overview, descriptive statistics, missing-value analysis, class-distribution analysis, and feature-correlation analysis.

4.1 Class Distribution

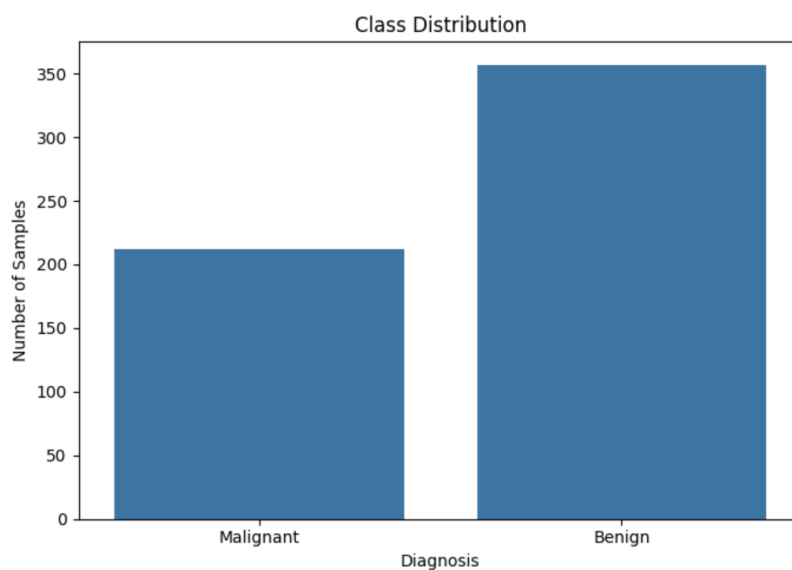


Figure 1: Class distribution of the WDBC dataset.

Inference: The two target classes have different frequencies. Stratified splitting and cross-validation preserve class proportions and make the model comparison more consistent.

4.2 Feature Correlation

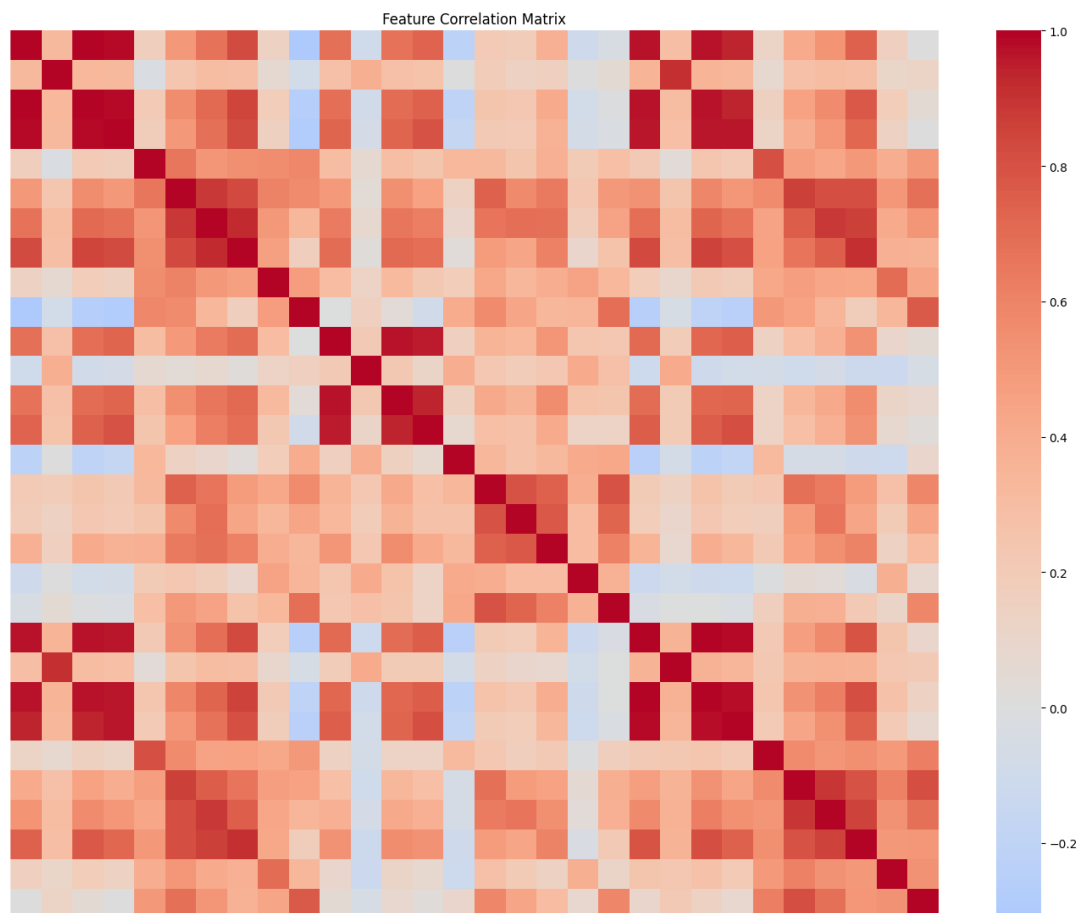


Figure 2: Correlation matrix of the 30 numerical features.

Inference: Several measurements are strongly correlated because they describe related properties of cell nuclei. Decision Trees and Random Forests do not require feature scaling. Random Forest additionally uses random feature subsets to increase tree diversity.

5 Data Preprocessing

- No missing-value imputation is required because the dataset contains no missing values in the supplied representation.
- The target is binary and is mapped to Malignant and Benign for interpretation.
- Feature scaling is not required for tree-based models.
- An 80–20 stratified train-test split is used.
- No additional feature engineering is applied.

6 Machine Learning Algorithm

6.1 Decision Tree Classifier

A Decision Tree recursively splits the feature space using impurity reduction. For class probabilities $p_1, \dots, p_K$, Gini impurity is

$$G = 1 - \sum_{k=1}^K p_k^2,$$

while entropy is

$$H = - \sum_{k=1}^K p_k \log_2(p_k).$$

A tree chooses splits that improve node purity. Greater depth permits more complex decision rules but can increase variance and overfitting.

Advantages: easy interpretation, no scaling requirement, and ability to model non-linear relationships.

Limitations: sensitivity to training-data changes, high variance, and overfitting when grown excessively deep.

6.2 Random Forest Classifier

Random Forest combines many decision trees trained using randomized observations and feature subsets. For classification,

$$\hat{y}(x) = \text{mode}\{\hat{y}_1(x), \hat{y}_2(x), \dots, \hat{y}_B(x)\}.$$

Bootstrap sampling and random feature selection reduce correlation among individual trees, while aggregation reduces variance.

Advantages: improved stability, reduced variance, nonlinear modelling, and strong generalization in many tabular settings.

Limitations: greater computational cost and reduced direct interpretability compared with one tree.

7 Experimental Setup

Python Version	Recorded by notebook at runtime
Scikit-Learn Version	Recorded by notebook at runtime
IDE	Jupyter Notebook / JupyterLab
Operating System	Recorded by notebook at runtime
Hardware	Recorded by notebook at runtime
Random State	42
Cross-Validation	Stratified 5-Fold
Test Size	20%

8 Hyperparameter Selection

8.1 Decision Tree

The search space includes criterion (Gini, entropy, log loss), maximum depth (2, 3, 4, 5, 7, 10, unrestricted), minimum samples split (2, 5, 10), and minimum samples leaf (1, 2, 4).

8.2 Random Forest

The search space includes number of estimators (50, 100, 200), maximum depth (unrestricted, 5, 10, 15), maximum features (sqrt, log2, all), and bootstrap (True/False).

The search uses stratified 5-fold cross-validation. Mean F1-score is used as the refit criterion, while mean accuracy is also recorded.

“latex

9 Hyperparameter Tuning Results

Table 2: Decision Tree Hyperparameter Evaluation using 5-Fold Cross-Validation

Criterion	Max Depth	Avg CV Accuracy (%)	Avg CV F1 Score (%)
Gini	5	93.85	95.06

Best parameters: {'criterion': 'gini', 'max_depth': 5, 'min_samples_leaf': 4, 'min_samples_split': 2}

Best mean CV F1: 0.95058

Table 3: Random Forest Hyperparameter Evaluation using 5-Fold Cross-Validation

n estimators	Max Depth	Max Features	Avg CV Accuracy (%)	Avg CV F1 Score (%)
100	None	log2	96.48	97.18

Best parameters: {'bootstrap': True, 'max_depth': None, 'max_features': 'log2', 'n_estimators': 100}

Best mean CV F1: 0.97180

10 5-Fold Cross-Validation Performance Comparison

Table 4: 5-Fold Cross-Validation Accuracy Comparison

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	0.9451	0.9451	0.9121	0.9231	0.9670	0.9385
Random Forest	0.9670	0.9780	0.9341	0.9560	0.9890	0.9648

11 Results

Table 5: Final Evaluation Metrics on the Held-Out Test Set

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Decision Tree	0.9035	0.9420	0.9028	0.9220	0.9358
Random Forest	0.9561	0.9589	0.9722	0.9655	0.9924

““

12 Result Visualization

12.1 Decision Tree Depth

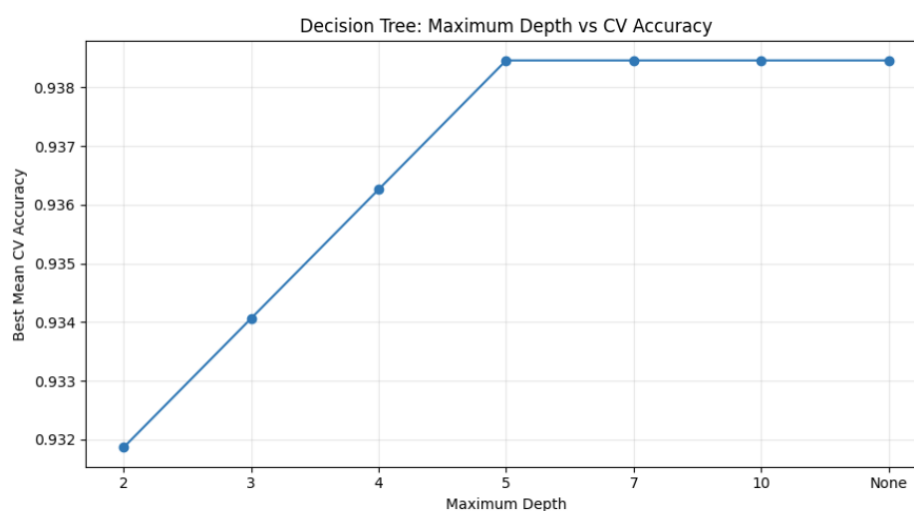


Figure 3: Maximum depth versus best cross-validation accuracy for the Decision Tree.

Inference: Depth controls tree complexity. Shallow trees can underfit, while excessive depth can increase variance. Cross-validation provides an empirical basis for selecting a suitable depth.

12.2 Decision Tree Structure

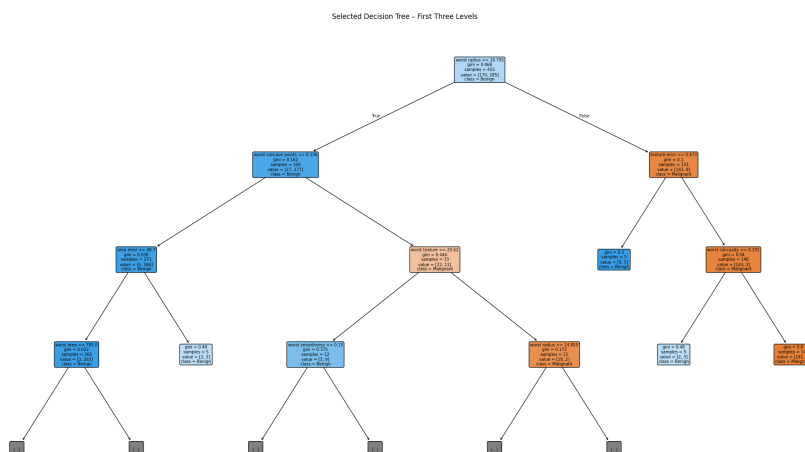


Figure 4: First three levels of the selected Decision Tree.

Inference: The tree shows how feature thresholds recursively divide the observations. The displayed first three levels provide an interpretable view of the learned decision rules.

12.3 Random Forest Number of Estimators

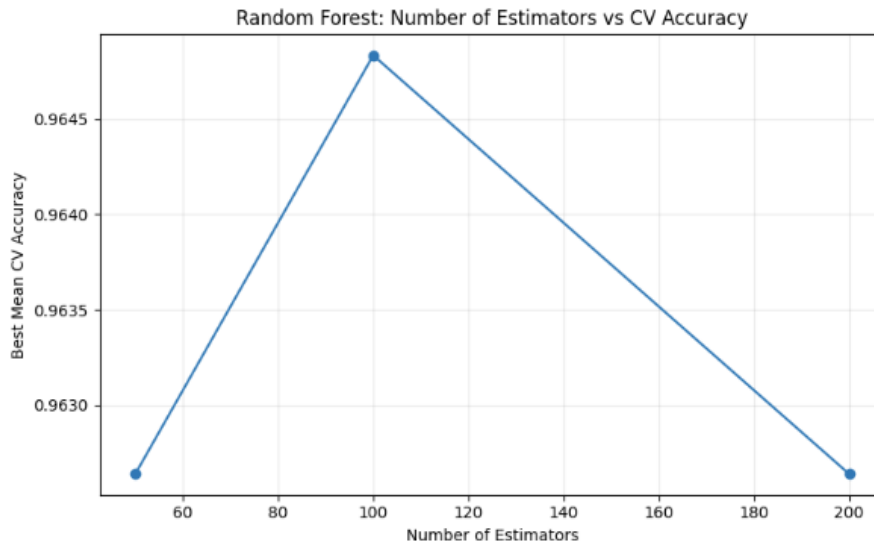


Figure 5: Number of Random Forest estimators versus best cross-validation accuracy.

Inference: More trees can stabilize predictions by aggregating more randomized learners. Performance may reach a plateau, producing diminishing returns for additional trees.

12.4 Decision Tree Confusion Matrix

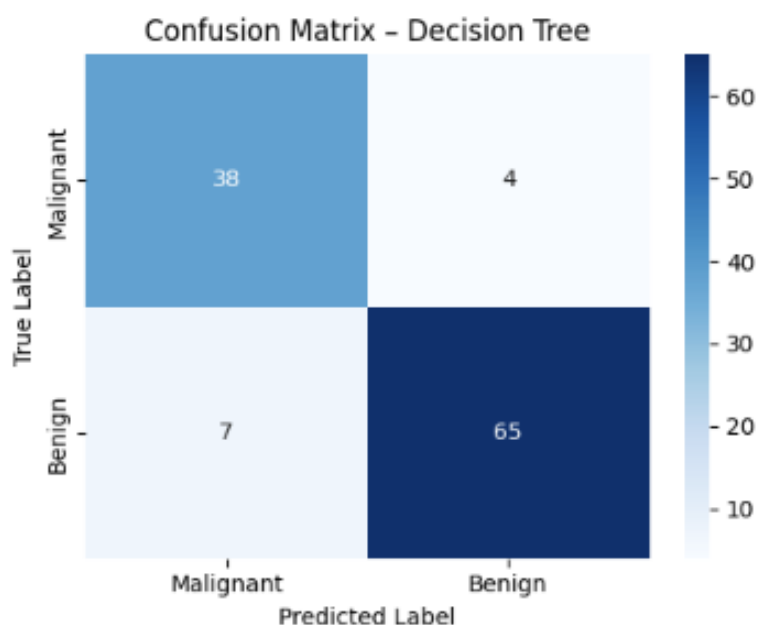


Figure 6: Confusion matrix for the selected Decision Tree.

Inference: The matrix shows correct and incorrect predictions by class. It complements aggregate metrics by revealing the distribution of false positives and false negatives.

12.5 Random Forest Confusion Matrix

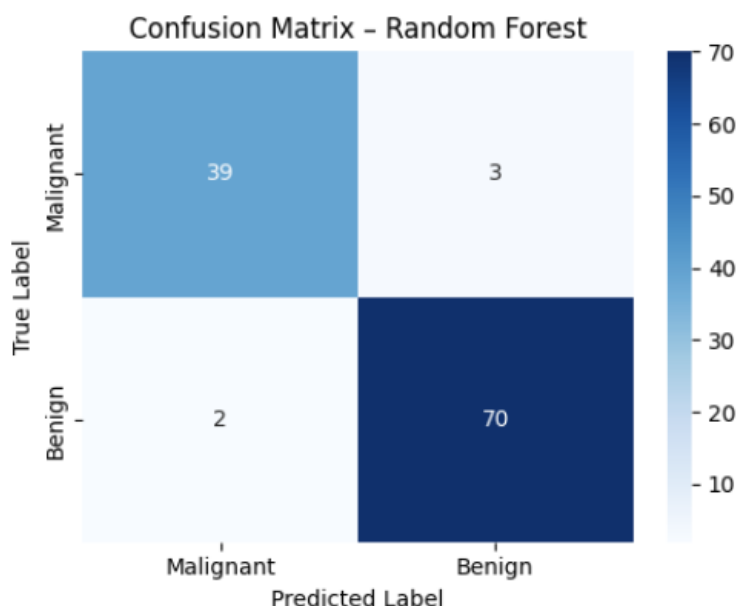


Figure 7: Confusion matrix for the selected Random Forest.

Inference: The Random Forest matrix provides class-specific information about its errors. Comparing it with the Decision Tree matrix shows how ensemble aggregation

changes prediction behaviour.

12.6 ROC Curve

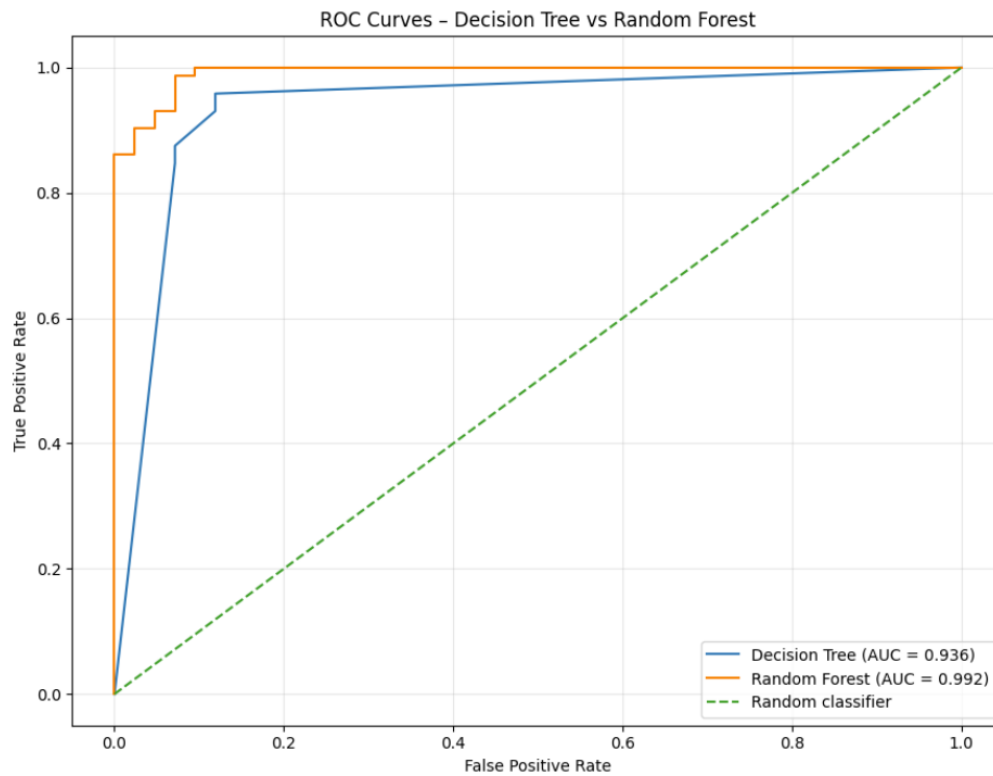


Figure 8: ROC curves comparing Decision Tree and Random Forest.

Inference: ROC curves show the trade-off between true-positive and false-positive rates across thresholds. AUC summarizes discrimination across those thresholds.

12.7 Random Forest Feature Importance

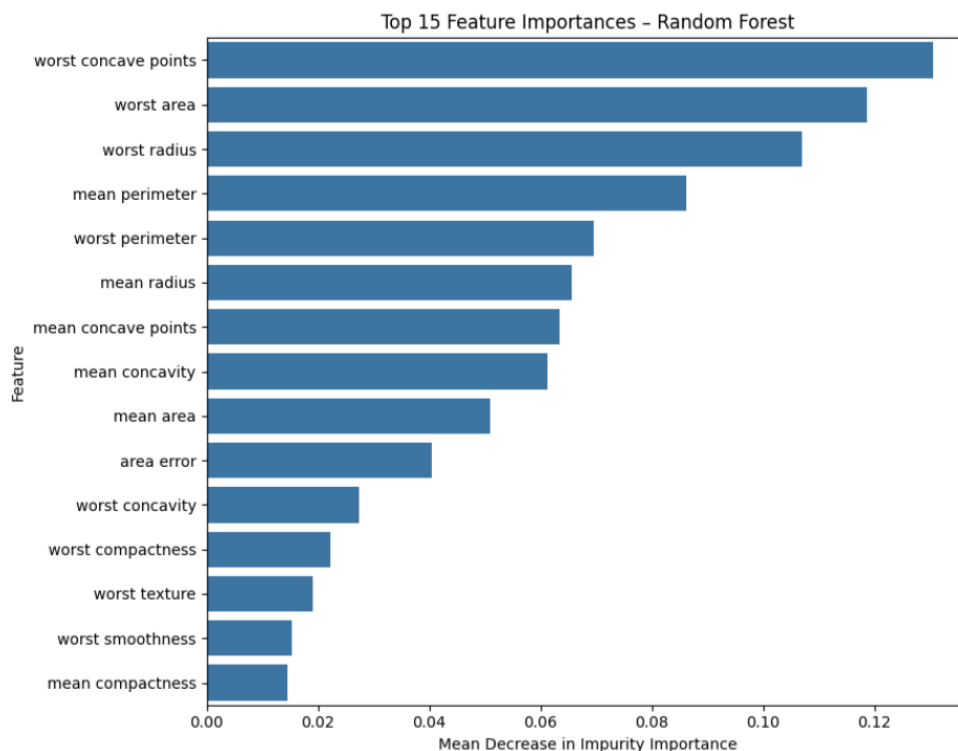


Figure 9: Top 15 Random Forest feature importances.

Inference: Feature importance identifies variables that contribute strongly to tree splitting in the fitted forest. These values are model-specific and do not imply causality.

13 Discussion

13.1 Effect of Tree Depth

Increasing depth allows increasingly complex decision rules. Training error can decrease as depth increases, but a very deep tree can model noise and increase variance. Cross-validation helps select a more appropriate complexity level.

13.2 Hyperparameter Impact

Decision Tree performance is influenced by criterion, maximum depth, minimum samples split, and minimum samples leaf. Random Forest performance depends on ensemble size, tree depth, feature subsampling, and bootstrap sampling. The GridSearchCV tables provide the empirical comparison of these settings.

13.3 Generalization

Random Forest improves stability by averaging many randomized trees. Bootstrap sampling gives trees different training samples, while random feature selection reduces cor-

relation among trees. This generally reduces variance relative to a single high-variance Decision Tree.

13.4 Does Ensemble Learning Always Improve Performance?

No fixed improvement should be assumed before observing the results. The ensemble may improve some metrics while not improving every metric on a particular finite test set. Dataset characteristics, tree diversity, hyperparameters, and sampling variation all influence the outcome.

14 Observation Questions

1. How does tree depth affect overfitting in Decision Trees?

Greater depth increases model complexity. Shallow trees may underfit, whereas very deep trees may fit noise and training-specific patterns, increasing variance. Cross-validation helps select a suitable depth.

2. Which hyperparameter had the greatest impact on performance?

This should be determined from the actual GridSearchCV results. Compare the validation scores across criterion, maximum depth, minimum samples split, and minimum samples leaf for the Decision Tree, and across number of estimators, maximum depth, maximum features, and bootstrap for Random Forest.

3. How does Random Forest improve generalization?

It combines many trees trained with bootstrap samples and random feature subsets. Aggregating diverse trees reduces dependence on any individual high-variance tree and can improve stability on unseen data.

4. Did ensemble learning always improve performance? Why or why not?

Use the measured cross-validation and test-set metrics to answer this. Ensemble learning does not guarantee improvement on every finite test set or every metric; its benefit depends on model diversity, hyperparameters, dataset structure, and sampling variability.

15 Conclusion

Decision Tree and Random Forest classifiers were implemented and evaluated on the Wisconsin Diagnostic Breast Cancer dataset. Hyperparameters were selected using stratified 5-fold cross-validation, and final performance was measured on an independent 20% test set. The experiment demonstrates how tree complexity affects overfitting and how Random Forest uses bootstrap sampling, random feature selection, and aggregation to reduce variance and improve stability.

16 References

- [1] Scikit-learn, “Decision Trees,” <https://scikit-learn.org/stable/modules/tree.html>.
- [2] Scikit-learn, “Ensemble Methods / Random Forest,” <https://scikit-learn.org/stable/modules/ensemble.html>.
- [3] UCI Machine Learning Repository, “Breast Cancer Wisconsin (Diagnostic) Dataset,” <https://archive.ics.uci.edu/>.
- [4] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.